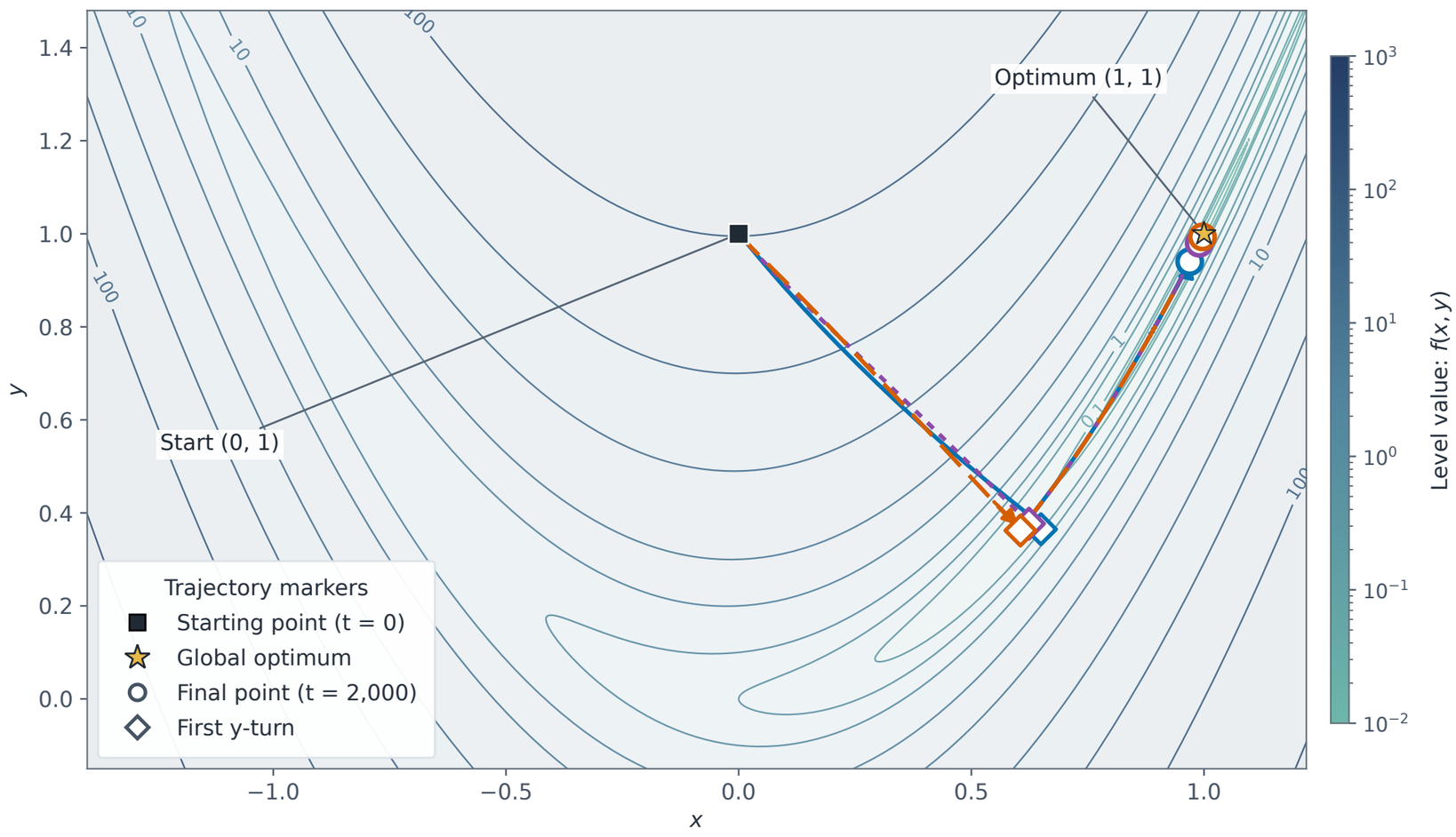
Rosenbrock: AdamW geometry, Hessian and gradient-identity kappa

2,000 updates per learning rate | $\beta_1 = 0.9$, $\beta_2 = 0.95$ | CPU float64

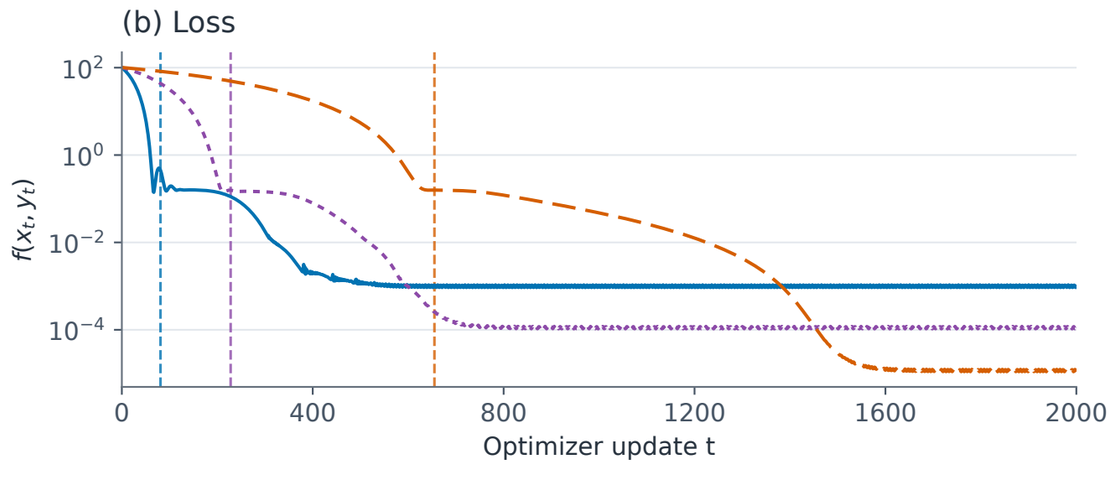
Learning rate | vertical dashed line at first y-turn

$\eta = 10^{-2}$ | turn t = 81 $\eta = 3.1623 \times 10^{-3}$ | turn t = 228 $\eta = 10^{-3}$ | turn t = 655

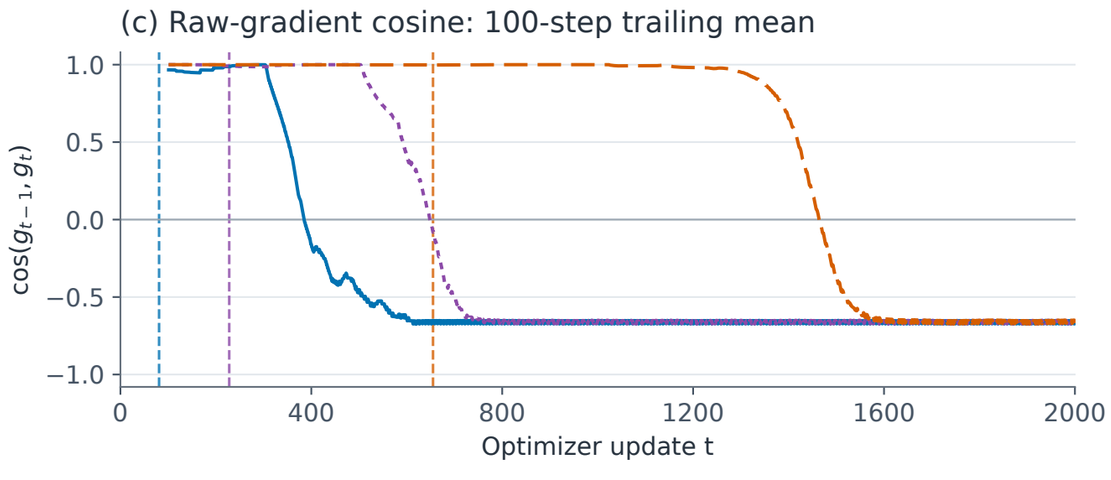
(a) Full optimization trajectories and objective level curves



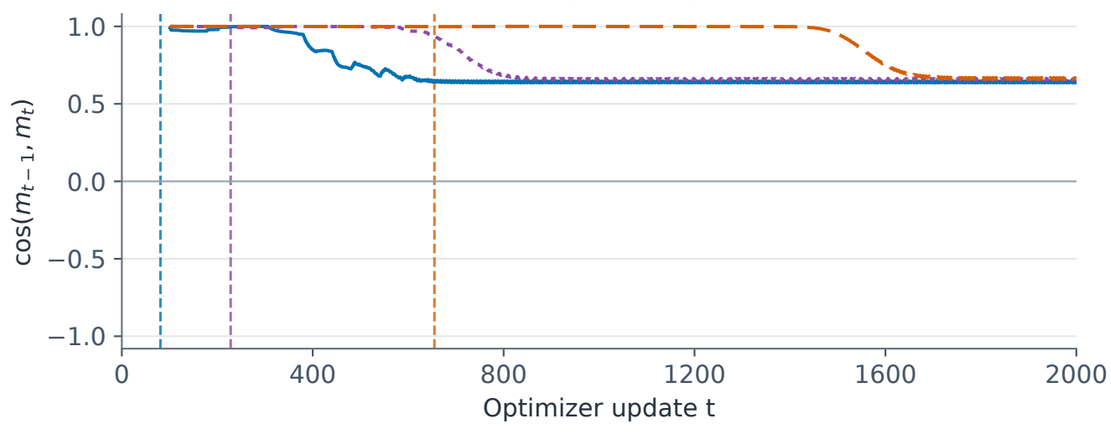
(b) Loss



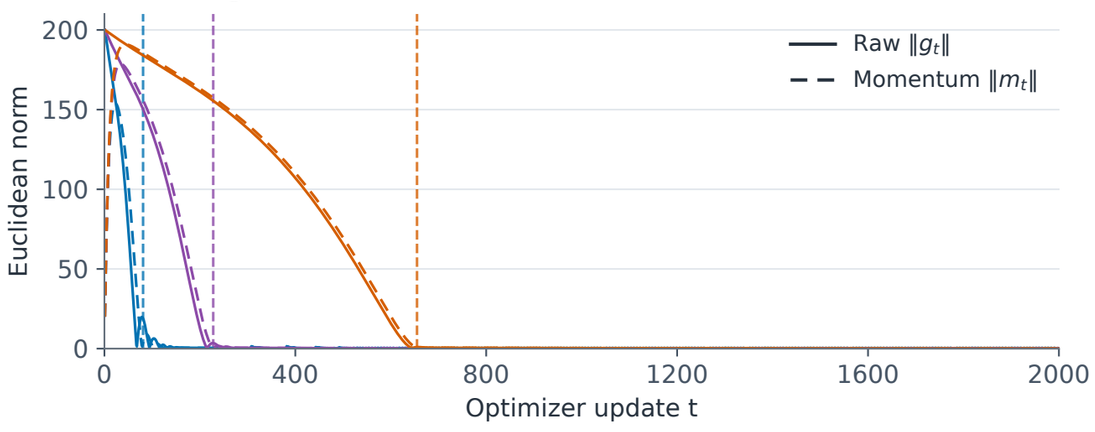
(c) Raw-gradient cosine: 100-step trailing mean



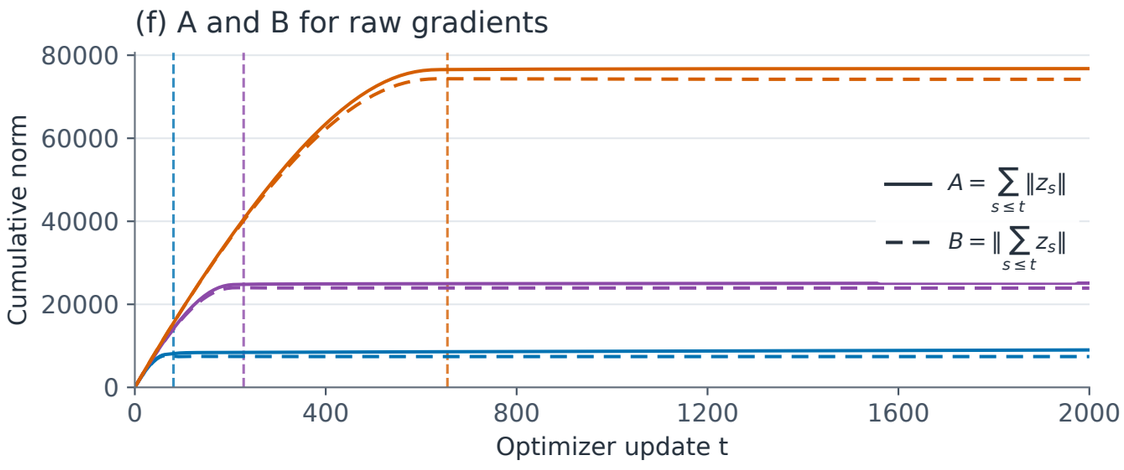
(d) Momentum cosine: 100-step trailing mean



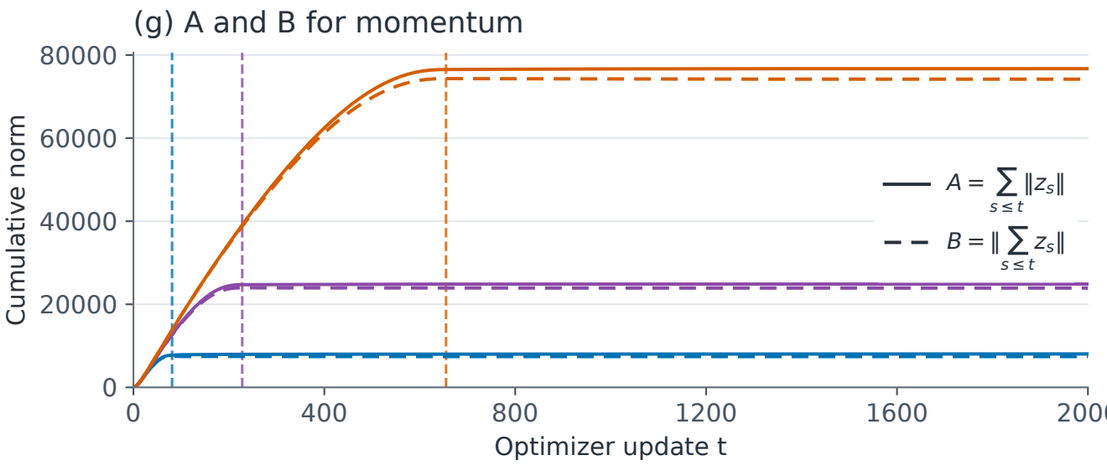
(e) Raw-gradient and momentum vector norms



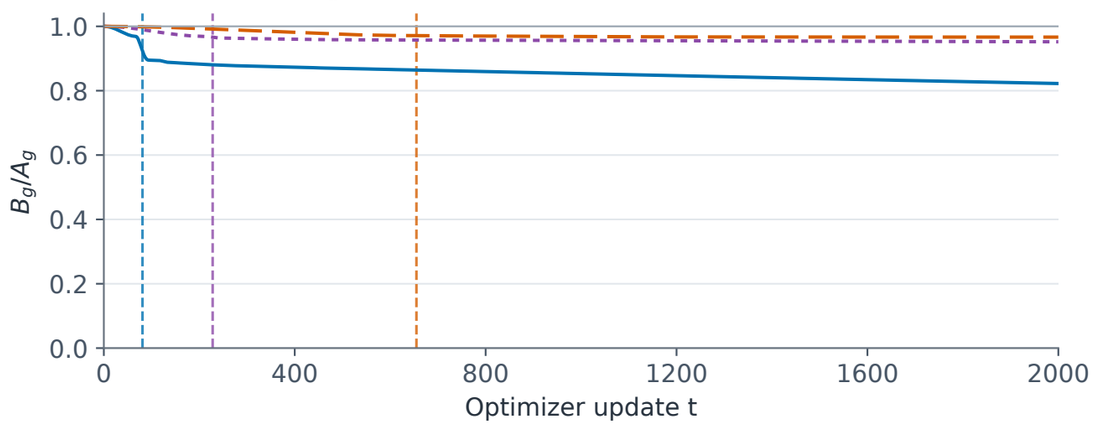
(f) A and B for raw gradients



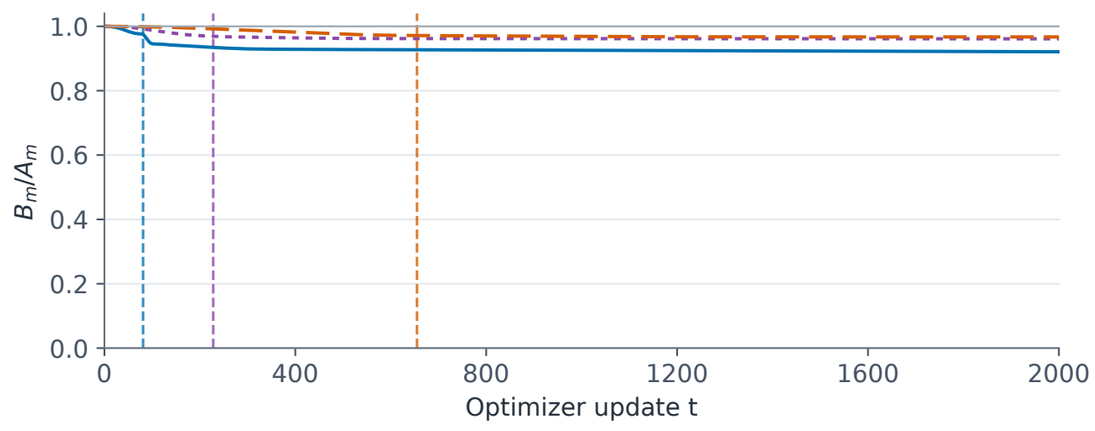
(g) A and B for momentum



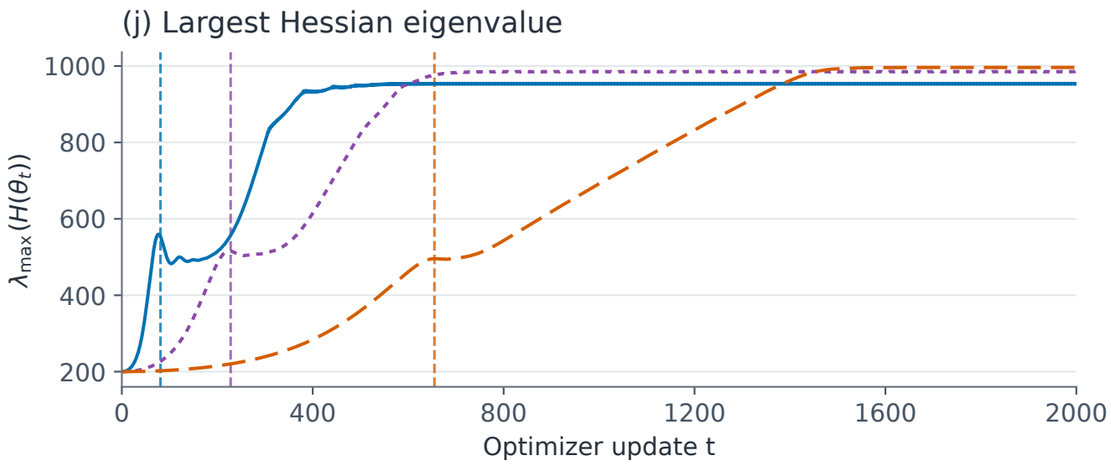
(h) B/A for raw gradients



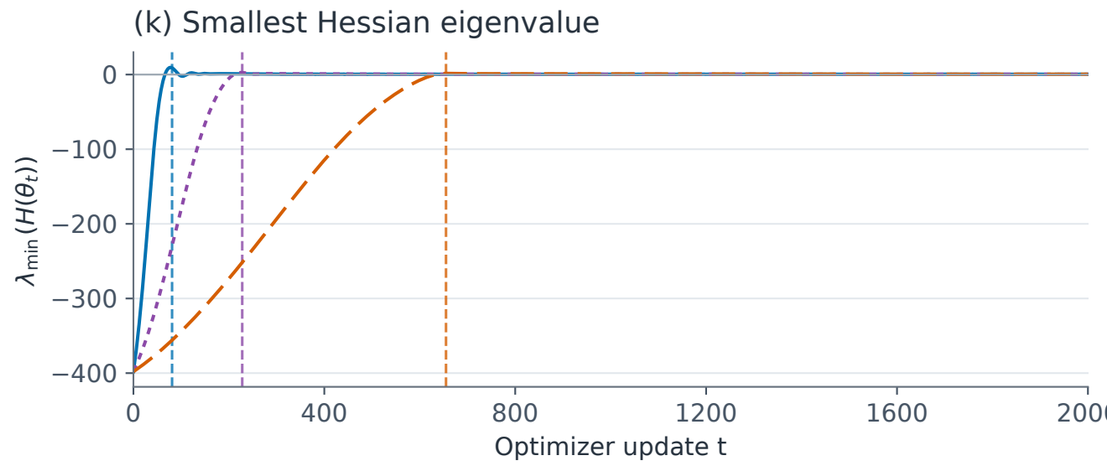
(i) B/A for momentum



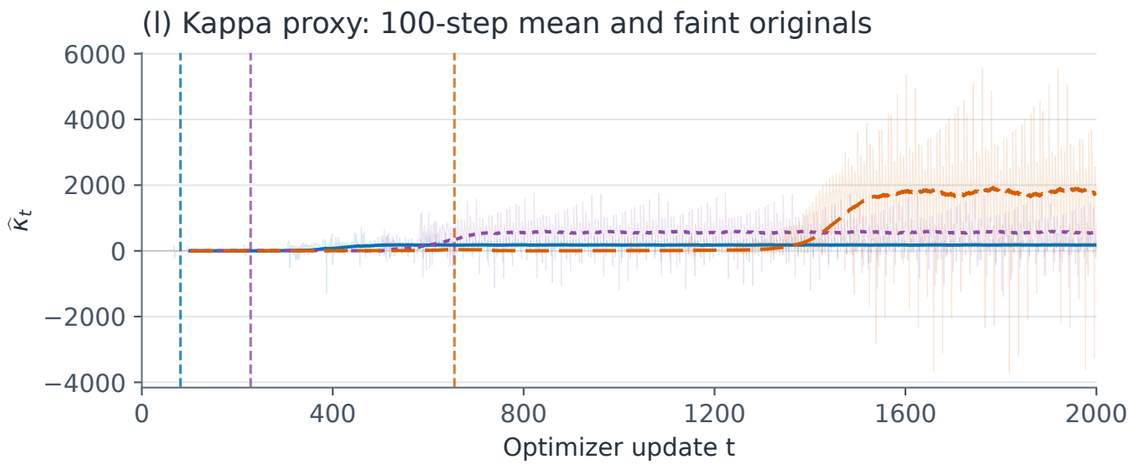
(j) Largest Hessian eigenvalue



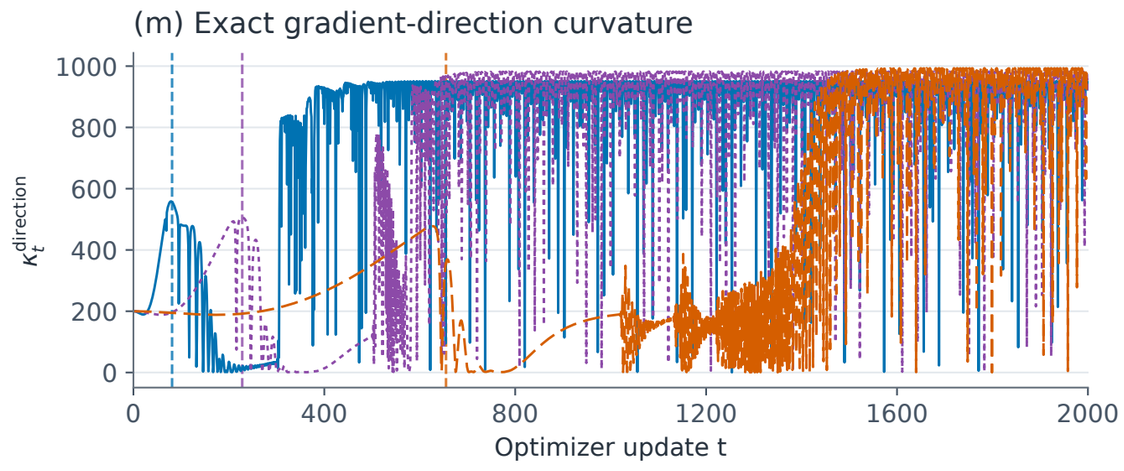
(k) Smallest Hessian eigenvalue



(l) Kappa proxy: 100-step mean and faint originals



(m) Exact gradient-direction curvature



$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$. Panels (c,d): 100-step trailing means; complete finite windows only. Panel (l) averages the unsmoothed identity; faint raw values remain.

m_t is the stored, uncorrected first moment. Diamonds and LR-colored dashed vertical lines mark the first y-turn.

Vectors at update t use $g_t = \nabla f(x_{t-1}, y_{t-1})$; loss uses (x_t, y_t). First adjacent cosines are undefined. B/A is undefined if A = 0.

Hessian panels: theta[t]. Kappa panels: pair t uses H(theta[t-2]); epsilon = 0. SGD identity evaluated as an AdamW proxy, not a stability threshold.